MIL-J-21994B(OS)
31 May 1978
Superseding
MIL-J-21994A(OS)
4 January 1967

MILITARY SPECIFICATION

JUNCTION BOX MARK 43 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement of a Junction Box Mark 43 Mod 0, LD 291892. The Junction Box Mark 43 Mod 0 is of one type and of one class.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident. 10001)

LD 291892	Junction Box Mark 43 Mod 0

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Md. 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1350

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All drawings referenced herein are listed on LD 291892 unless otherwise designated.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OD 10511	Nonmagnetic Testing with the Test Set Mark 115 Mod 4 & 5

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

HANDBOOK H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

3. REQUIREMENTS

3.1 Description. The junction box shall be a rectangular metal box with a cover which contains five cable receptacles: four 35-pin and one 4-pin receptacles.

3.2 First article sample. Unless otherwise specified in the contract or order (see 6.2), a first article sample of six junction boxes shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be packaged in accordance with section 5 herein and tested as specified in section 4 herein. The sample is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing items that comply with this specification and the design and may be waived by the contracting officer if satisfactory production has been recently accomplished.

3.3 General requirements.

3.3.1 Fabrication. The Junction Box Mark 43 Mod 0 shall be manufactured in accordance with LD 291892, all documents listed therein and as specified herein. Inspection and acceptance shall be in accordance with section 4.

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3.3.2 Interchangeability. Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in accordance with applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection for fit or performance.

3.3.3 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.3.4 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

3.4 Performance requirements and product characteristics. The junction box shall meet the following performance requirements and product characteristics:

3.4.1 Electrical continuity. The resistance between all points of the electrical circuits connected by design shall not exceed 0.5 ohm when tested in accordance with 4.7.3.

3.4.2 Insulation resistance. The electrical resistance between each terminal of the junction box and every other terminal insulated from it by design, and between each terminal and the junction box case, shall be 50 megohms minimum when measured with an applied potential of 500 ± 50 volts DC and tested in accordance with 4.7.4.

3.4.3 Static magnetic effect. The static magnetic effect of the Junction Box Mark 43 Mod 0 shall not cause a total change in the flux density of the background magnetic field of more than 0.10 milligauss (peak to trough).

3.5 Environmental resistance. The Junction Box Mark 43 Mod 0 shall withstand the following hazards of its functioning environment:

3.5.1 Vibration. The Junction Box Mark 43 Mod 0 shall withstand the vibration test of 4.8.1 without physical damage and thereafter meet the requirements of 3.4.1 and 3.4.2.

3.5.2 Shock. The Junction Box Mark 43 Mod 0 shall withstand the shock test of 4.8.2 without physical damage and thereafter meet the requirements of 3.4.1 and 3.4.2.

3.5.3 Temperature. The Junction Box Mark 43 Mod 0 shall withstand the temperatures given below during operation and storage.

3.5.3.1 Operating. The Junction Box Mark 43 Mod 0 shall meet the requirements of 3.4.1 and 3.4.2 when tested according to 4.8.3.1.

3.5.3.2 Storage. The Junction Box Mark 43 Mod 0, packaged and packed as required by Section 5 shall be able to withstand the test of 4.8.3.2 without damage. After being removed from the chamber, the junction box shall meet the requirements of 3.4.1 and 3.4.2.

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3.6 Cleanliness. Necessary precautions shall be taken to protect the junction box components from moisture, solder splashes, chips and other foreign material during the assembly process.

3.7 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the item covered by this specification shall be classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
c. Periodic production testing (see 4.6)

4.3 Test equipment. The following items of test equipment are necessary to perform the tests required in this specification:

4.3.1 An ohmmeter having a range of 0.0 to 1.0 ohm and an accuracy of ± 1.0 percent.

4.3.2 A resistance measuring device with 500 ± 50 volts D-C potential, capable of measuring at least 50 megohms at ± 5 percent accuracy.

4.3.3 Test set Mark 115 type with Test Stand Mark 9 Mod 0 and Adapter Mark 31 type (Government furnished).

4.4 First article inspection. First article sample approval will be granted when the Government testing activity, through testing and subsequent analysis of the test results, is assured of contractor conformance to the contract technical requirements. Conditional acceptance may be granted at the option of the Government based on analysis of detected defects. A first article sample may be disapproved when either nonconforming characteristics or unauthorized materials are found. The tests of 4.7 and 4.8 will be performed in the sequence and with the sample quantities given in Figure 1.

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TABLE I. Correlation of requirement and test method paragraphs.

Title of Examination or Test	Requirement Paragraph	Method Paragraph	Inspection Classifications Quality (1) Conformance	First Article & (2) Periodic Production
Group A				
Packaging	5	4.7.1	X	X
Visual Examination	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.6 and 3.7	4.7.2	X	X (3)
Continuity	3.4.1	4.7.3	X	X
Insulation resistance	3.4.2	4.7.4	X	X
Static magnetic effect	3.4.3	4.7.5	X	X
Group B				
Temperature	3.5.3	4.8.3		
Operating	3.5.3.1	4.8.3.1		X
Storage	3.5.3.2	4.8.3.2		X
Group C				
Vibration	3.5.1	4.8.1		X
Shock	3.5.2	4.8.2		X

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TABLE I. (continued)

Notes

(1) Quality conformance inspection shall be an in-process function performed during assembly of the junction boxes prior to preparation for delivery by the manufacturer.

(2) See Figure 1 for test sequences for the first article and periodic samples which shall be performed by a designated Government testing activity.

(3) Do not open the junction boxes for this examination until the continuity and insulation resistance tests have been performed.

(4) The X in each column indicates that the tests are to be performed as appropriate in accordance with notes (1) and (2).

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4.5 Quality conformance inspection. Each production unit shall be subjected to the examination of 4.7.2 and tests designated as 100 percent in 4.7 (see Table I). Any unit containing one or more defects shall be cause for regarding the unit as defective.

4.6 Periodic production testing. After the quality conformance inspection, production shall be sampled periodically for additional testing, as shown in Table I and Figure 1, at a designated Government activity.

4.6.1 Periodic sampling. Each month's production of junction boxes, unless otherwise specified, shall be designated as a lot for acceptance sampling. Six units shall be selected at random from each lot by the Government inspector.

4.6.2 Test result. Failure of any periodic production sample to meet all of the specification, drawing, and contract requirements as a result of the test designated in Table I, may result in rejection of the lot.

4.7 Examination and test methods.

4.7.1 Packaging. The packaging shall be examined to determine conformance to section 5 herein.

4.7.2 Visual examination. Each Junction Box Mark 43 Mod 0 shall be visually examined in accordance with the drawing 1378506 to determine compliance with this specification and its associated drawings as specified in 3.3.1. Examination in accordance with drawing 1378506 shall not relieve the contractor of his responsibility to meet all requirements of other drawings listed thereon.

4.7.3 Continuity test (100%). Using the ohmmeter, item 4.3.1, measure the resistance of each electrical circuit of the junction box between directly connected terminals. Verify that the resistance of every such circuit does not exceed 0.5 ohm.

4.7.4 Insulation resistance test (100%). Place the junction box within a conditioning chamber. Maintain the junction box for 24 hours at a relative humidity of 75 ± 5 percent and a temperature of 77°F ± 2°F. Within 15 minutes after removal of the junction box from the chamber, measure the insulation resistance between each terminal of the junction box and every other terminal not electrically connected by design, and between each terminal and the junction box case, using the resistance measuring device, item 4.3.2. Verify that the insulation resistance is not less than 50 megohms. To facilitate production, the contracting officer may allow the insulation resistance test of this paragraph to be conducted at ambient temperature and relative humidity without previous conditioning. However, if the junction box fails the test at ambient conditions, it shall be conditioned and tested in accordance with the procedure specified above.

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4.7.5 Static magnetic effect test (100%). Test the Junction Box Mark 43 Mod 0 in accordance with Test Method II of OD 10511, under the general test conditions and definitions described therein. Mount the junction box on the Adapter Mark 31 Mod 0 on the Test Stand Mark 9 Mod 0. The test face of the junction box is the largest flat area face near the four-contact connection, J12. The test distance is five inches. To achieve this test distance, the adapter shall be located by adjusting the adapter plate until the indicators are positioned at the 10 ± 1/16 inch marks on the Test Stand Mark 9 Mod 0. The test sample shall meet the requirement of 3.4.3 during this test.

4.8 Environmental tests.

4.8.1 Vibration. The junction box shall be rigidly fastened directly to the vibration table by either simulating the actual method of mounting used in service or by its external case, whichever is appropriate. Simple harmonic excitation shall be applied parallel to each of the three principal axes of the junction box for equal periods of time. The frequency range of 10 to 60 Hz may be covered either by slowly cycling at a logarithmic rate between the limits, or by at least 24 discrete frequency steps which have a logarithmic distribution. The vibration amplitude shall be maintained at 2.0 ± 0.2 g vector or 0.10 ± 0.01 inch, whichever is the lesser. Duration of the test in each direction is four hours. The total test duration shall be 12 hours. The sample shall meet the requirements of 3.5.1 after this test.

4.8.2 Shock. The Junction Box Mark 43 Mod 0 shall be subjected to a continuous two-phase shock pattern acting in a direction parallel to the top and front faces of the box. The samples shall be shock-tested at a temperature of -65°F. The mounting of the samples in the test vehicle shall simulate service conditions. The test pattern shall be as follows:

a. The first (impact) phase shall consist of a velocity change of 5.5 to 7 feet per second in an elapsed time of 0.2 to 0.4 millisecond.

b. During the second (drag) phase, the junction box shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time the minimum acceleration is 90 gravity units. The peak acceleration shall be between 125 and 150 gravity units.

The sample shall meet the requirements of 3.5.2 after this test.

4.8.3 Temperature tests. The sample will be tested for physical integrity and operability as specified below:

4.8.3.1 Operating. Subject the two sample junction boxes to the quality conformance tests of 4.7.3 and 4.7.4, one each at 28° ± 2°F and 90° ± 3°F(see Figure 1). Verify that the requirements of 3.5.3.1 are met.

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4.8.3.2 Storage. Subject the junction boxes, packaged and packed as specified in section 5, to one 14-day TEMPERATURE AND HUMIDITY CYCLE defined in MIL-STD-331, Test Method 105.1, except that relative humidity in the chamber need not be raised above room ambient or ambient as controlled. The sample shall then be tested for conformance to 3.4.1 and 3.4.2 at ambient conditions.

5. PACKAGING

5.1 Preservation-packaging and packing. The Junction Boxes Mark 43 Mod 0 shall be packaged and packed in accordance with LD 299621.

NOTE: When junction boxes are purchased for a prime contract for installation into Assembly Condition "D" mines, preservation-packaging and packing shall be adequate to protect the unit prior to installation and the requirements of LD 299621 do not apply.

5.2 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Junction box is intended for use in underwater mines.

6.2 Ordering data. Procurement documents should include the following:

a. Designated Government activity to perform first article and periodic production sample tests.

b. When first article samples are waived because of having been submitted on a recent contract.

c. Any periodic production test sampling at variance with this specification, with a new test plan to designate the desired level and sequence of testing for each of the sample units. The allowable test failure rate should also be designated.

d. Special test equipment to be loaned by the Government and the terms under which it is loaned.

e. Title, number and date of this specification.

f. When junction boxes are purchased for a prime contract for installation into Assembly Condition "D" Mines.

g. Lot size if other than specified (see 4.6.1).

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6.3 Definitions. The following terms are used throughout this specification for the Junction Box Mark 43 Mod 0 and are herein defined for clarity.

6.3.1 Junction Box. Reference herein to the junction box, assembly, or unit shall be understood to mean the Junction Box Mark 43 Mod 0.

6.3.2 Lot. Unless otherwise specified in the contract or purchase order, a lot size shall be one month's scheduled production.

6.4 The design and technical cognizance for the junction box specified herein are located at the Naval Mine Engineering Facility, Yorktown, VA 23691.

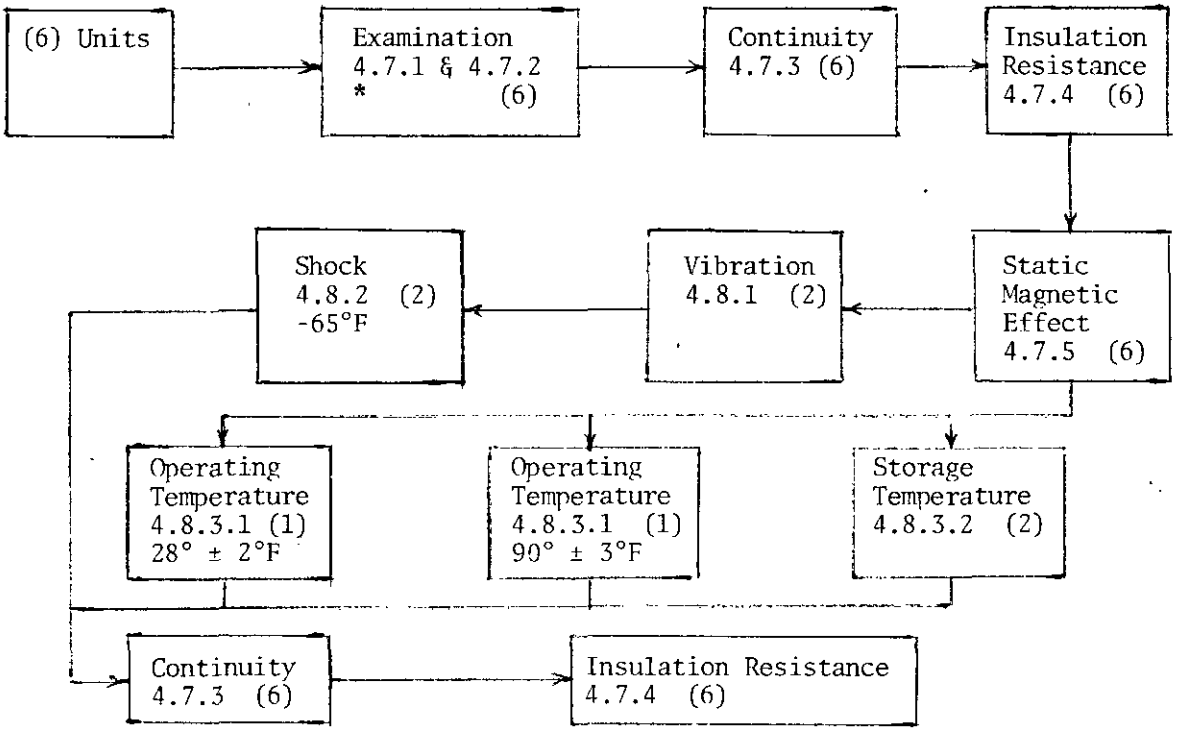
6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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*See notes under Table I.

The number in parenthesis in each block indicates the quantity to be tested at that point.

FIGURE I. First article and periodic production test sequence.

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MIL-J-21994B(OS) "JUNCTION BOX MARK 43 MOD 0"

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